

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures described below in Section 3.2.

After these restrictions we have records on between 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two different approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#). [Eloundou et al. \(2024\)](#) estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They then construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second primary approach we take uses data on generative AI usage from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on a sample of several million conversations with Claude, Anthropic’s generative AI model. It then aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it also reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this information as an estimate of whether usage of AI for an occupation is primarily complementary or substitutable with labor.¹⁴

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the Quarterly Census of Employment and Wages (QCEW) using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴Specifically, [Handa et al. \(2025\)](#) first use Claude to classify conversations into six categories: Directive, meaning complete task delegation with minimal interaction; Feedback Loop, meaning task completion guided by environmental feedback such as when repeatedly relaying coding errors to the model; Task Iteration, meaning a collaborative

whether usage of AI for an occupation is primarily a substitute or complement for labor.¹⁵ Table A1 shows example occupations that are in the highest and lowest exposure category for each measure.

Figure 6 shows employment changes by overall prevalence of related Claude queries. The patterns match the findings using the Eloundou et al. (2024) measures closely. Figure 7 likewise shows that the occupations with the highest estimated automation shares have experienced declining employment for the youngest workers.

In contrast, Figure 8 indicates that the occupations with the highest estimated augmentation shares have *not* experienced a similar pattern. Employment changes for young workers are not ordered by augmentation exposure, as the fifth quintile has among the fastest employment growth. The findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.¹⁶

4.4 Fact 4: Employment declines for young, AI-exposed workers remain after conditioning on firm-time shocks

While our results so far are consistent with the hypothesis that generative AI is causing a decline in entry-level employment, there are plausible alternative explanations. One class of explanations is that our patterns are explained by industry- or firm-level shocks correlated with sorting patterns by age and measured AI exposure. For example, one possibility is that young workers with high measured AI exposure are disproportionately likely to sort to firms heavily susceptible to interest rate increases.

We test for a class of such confounders by controlling for a rich set of fixed effects. For each age group, we estimate the Poisson regression

$$\log(E[y_{f,q,t}]) = \sum_{q' \neq 1} \sum_{j \neq -1} \gamma_{q',j} 1\{t = j\} 1\{q' = q\} + \alpha_{f,q} + \beta_{f,t} + \epsilon_{f,q,t} \quad (4.1)$$

¹⁵Figures A2 and A3 show that automation and augmentation results are similar when dropping occupations with low overall Claude usage.

¹⁶Note that occupations in the first two quintiles of the augmentation measure have very low Claude usage overall, with the average occupation in these quintiles comprising 0.01% and 0.09% of conversations, respectively. These occupations have a high share of conversations that are classified as neither automative nor augmentative. In contrast, occupations in the third through fifth quintiles average 0.47%, 0.39%, and 0.33% of Claude conversations. For the automation measure, overall Claude usage increases on average with the automation share, with the lowest exposure group averaging 0.05% of conversations and the highest group averaging 0.73%.

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁷

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁸ Standard errors are clustered by firm.

Results are in Figure 9, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁷Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁸Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.